Dongsu Song

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RESEARCH INTERESTS

- **Adversarial robustness:** Finding and Understanding adversarial vulnerabilities of vision systems, building on black-box sparse attacks and extending to video, 3D, multimodal foundation and VLA models.
- **Generative failure discovery:** Actively searching for semantic edge cases with generative models, understanding why vision models fail on them, and using these insights to make models more robust.
- **Visual privacy:** Repurposing adversarial perturbations to protect images and videos against unauthorized generative editing, image animation, and 3D/4D reconstruction.

EDUCATION

Korea Aerospace University, Republic of Korea

Ph.D. Student in Computer Science | Advisor: Prof. Jay Hoon Jung — *Mar. 2025 – Present*

M.S. in Artificial Intelligence | ***GPA: 4.50/4.50*** | *Advisor: Prof. Jay Hoon Jung* — *Mar. 2023 – Feb. 2025*

B.S. in Mechanical Engineering | ***GPA: 4.24/4.50 (Rank: 10/189)*** — *Mar. 2017 – Feb. 2023*

- **Magna Cum Laude**

PUBLICATIONS

Conference Publications

- **Dongsu Song**, Daeyun Go, Boseung Seo, and Jay Hoon Jung. "SegPAR: Class-Centric Decision-Based Sparse Attack for Semantic Segmentation." ***European Conference on Computer Vision (ECCV)***, 2026.
- **Dongsu Song**, Daehwa Ko, and Jay Hoon Jung. "Amnesia as a Catalyst for Enhancing Black Box Pixel Attacks in Image Classification and Object Detection." ***Advances in Neural Information Processing Systems (NeurIPS)***, 2024.

Workshop Proceedings

- **Dongsu Song**, Daeyun Go, Sunok Kim, and Jay Hoon Jung. "Prompt-Driven Generative Variations as EOT Views for Transferable Sparse-PGD Attacks." *Int. Workshop on Advanced Image Technology (IWAIT)*, Proc. SPIE, 2027.

Journal Articles

- **Dongsu Song** and Jay Hoon Jung. "A Quantitative Comparison of LIME and SHAP using Stamp-Based Distance Method on Image Data." *Journal of KIISE*, 50(10), 906–911, 2023.

Preprints and Submissions

- **Dongsu Song**, Daeyun Go, and Jay Hoon Jung. "Decision-Based Sparse Attacks on Structured Visual Predictions: Fixed-Reference Feedback from Segmentation to Detection." *Under review*, 2026.
- **Dongsu Song**, Daeyun Go, and Jay Hoon Jung. "Density Feedback Control for Transferability of Generative Sparse Adversarial Attacks." *Under review*, 2026.
- **Dongsu Song**, Daeyun Go, and Jay Hoon Jung. "Discovering Natural Transformation Vulnerabilities in Black-Box Vision Models." *arXiv:2609.07110, Under review*, 2026.

RESEARCH EXPERIENCE

Decision-Based and Generative Sparse Adversarial Attacks

Ph.D. Researcher, Quantum AI (QAI) Laboratory, Korea Aerospace University — *Mar. 2025 – Present*

- Developed SegPAR (ECCV 2026), a decision-based sparse attack with class-centric exploration and a discrepancy reward that tracks cumulative attack progress.
- Under a 1,000-query budget, lowered mIoU by 12.1 pp (CNNs) and 9.4 pp (Transformers) more than the strongest black-box baseline, averaged over all benchmarks, with the fewest perturbed pixels.
- Extended SegPAR from semantic segmentation to object detection with fixed-reference feedback (under review).
- Combined prompt-driven FLUX.2 edits with Gaussian EOT for Sparse-PGD, raising mean transfer ASR by 2.4 pp over Gaussian-only EOT, with gains on every CNN, Transformer, and SSM source (IWAIT 2027).
- Found that fixed sparsity coefficients stall transferability in generative sparse attacks by discarding effective pixels; proposed Density Feedback Control (DFC), which adapts the coefficient to the observed density (under review).
- Applied DFC to prior generator-based attacks, increasing mean transfer ASR by 9.1-15.6 pp (CNNs) and 4.4-5.2 pp (Transformers) with fewer pixels, reaching the target density.

Query-Efficient Black-Box Generation of Natural Adversarial Examples
Ph.D. Researcher; Advanced GPU Utilization Support Program, funded by MSIT *Apr. 2026 – Jul. 2026*

- Proposed the project and wrote the winning proposal, securing dedicated NVIDIA B200 GPU × 8 access for the lab.
- Developed Adversarial Scenario Attack (ASA), a query-based black-box framework combining a multimodal LLM and text-guided image editing with winner–loser feedback and greedy scenario composition.
- Under a 100-query budget, raised mean ASR by 43.9 pp over the strongest prior query-based attack across ten classifiers (CNNs, Transformers, SSMs, VLMs), and image-level transfer ASR by 26.4-29.9 pp.
- Showed cross-architecture transfer: "change lighting to dusk," from ResNet-50, raised error rates on all 13 CNN, Transformer, SSM, and VLM models by 10.3 pp on average over a control edit, reused unchanged on ImageNet classes.

Robust, Verified, and Trusted AI in Aviation
Graduate Researcher, Korea Aerospace University *Mar. 2023 – Aug. 2025*
Airbus Institute for Engineering Research (AIER), an Airbus-funded collaboration with USC

- Led development of RFPAR (NeurIPS 2024), an RL-guided sparse black-box attack for classification and detection.
- Designed a CNN-based agent and an Amnesia mechanism that reinitializes the agent and memory when rewards converge, retaining the best perturbed image.
- Raised mean ASR by 11.9 pp and cut mean queries by 25.8% relative to prior pixel attack across six ImageNet classifiers.

Adversarial Robustness under Quantization
Graduate Researcher; Industry Innovation Talent Program, funded by KIAT *Mar. 2023 – Dec. 2024*

- Proposed and led a 3-person study on whether adversarial robustness survives quantization in CNNs.
- Found that quantization reduced the robustness of adversarially trained CNNs, suggesting that robustness gains may not survive model compression.

Quantitative Evaluation of Visual Explanations
Student Researcher; Basic Research Program, funded by NRF *Mar. 2022 – Feb. 2023*

- Applied LIME and SHAP to diagnose misclassifications in satellite imagery and found that their explanations were inconsistent, motivating a quantitative comparison.
- Adapted a stamp-based distance method for quantitative LIME/SHAP comparison on controlled MNIST (KIISE 2023).

PROFESSIONAL ACTIVITIES

Invited reviewer: NeurIPS 2025; NLDL 2027.

Invited poster on RFPAR (NeurIPS 2024), Korean Conference on Computer Vision (KCCV) *Aug. 2025*

Mentoring, Korea Aerospace University *Spring 2023 – Fall 2025*

- Led ML fundamentals seminars for junior lab members; mentored undergraduates in an industry–academia project.

AWARDS AND SCHOLARSHIPS

- **Honorable Mention, 21st World Embedded Software Contest** (Korea) *Dec. 2023*
- **Research Excellence Scholarship** (Journal of KIISE) *Fall 2023*
- **Advisor-Selected Graduate Scholarship** (half tuition; sole recipient in lab) *Fall 2023 – Fall 2024*
- **Academic Excellence Scholarship** (half tuition; 2nd-highest GPA in department) *Spring 2021, Fall 2017*
- **Academic Merit Scholarship** (one-sixth tuition; top 10% in department) *Spring 2022, Fall 2021, Spring 2019*

MILITARY SERVICE

Republic of Korea Army *Jul. 2018 – Feb. 2020*
Squad Leader *Jan. 2019 – Jan. 2020*

- Led a 6-person squad and received the Division Commander's Award and Battalion Commander's Commendation.

SKILLS

- **Programming and tools:** Python, C, Git, Docker
- **Frameworks and libraries:** PyTorch, TensorFlow, Hugging Face, timm, MMSegmentation